

23103150022

Bihar Engineering University, Patna
B.Tech 3rd Semester Examination, 2024

Course: B.Tech
Code: 100302

Subject: Analog Electronic Circuits

Time: 03 Hours
Full Marks: 70

Instructions:-

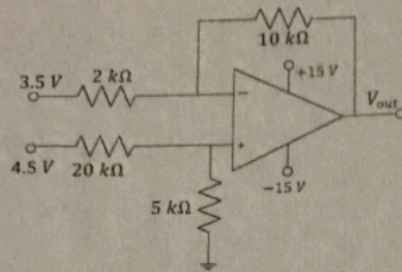
- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven questions only):-

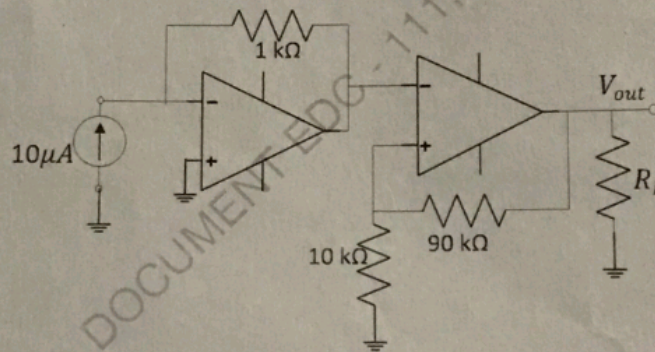
[2 × 7 = 14]

- (a) A Zener diode rated 5.1V is used with a 220Ω resistor. If input is 12V, what is the Zener current?
(i) 31.4 mA (iii) 20 mA
(ii) 50 mA (iv) 10 mA
- (b) At room temperature, a transistor has a collector current of 5 mA and $V_{BE} = 0.7V$. Find g_m .
(i) 7.14 S (iii) 5 S
(ii) 194 mS (iv) 3.5 S
- (c) Which region does a MOSFET operate in for amplification?
(i) Cutoff (iii) Saturation
(ii) Triode (iv) Ohmic
- (d) The trans-conductance (g_m) of a MOSFET is a measure of:
(i) Voltage gain (iii) Output conductance
(ii) Input impedance (iv) Change in drain current per unit change in gate voltage
- (e) Which parameter limits the maximum frequency response of an op-amp?
(i) Slew rate (iii) Input bias current
(ii) Offset voltage (iv) Common-mode gain
- (f) An inverting amplifier has $R_f = 100\text{ k}\Omega$ and $R_{in} = 10\text{ k}\Omega$. What is the voltage gain?
(i) -100 (iii) 10
(ii) -10 (iv) 1
- (g) Which op-amp circuit converts a sinusoidal input to a square wave output?
(i) Peak detector (iii) Schmitt trigger
(ii) Comparator (iv) Integrator
- (h) A MOSFET has $I_D = 2\text{ mA}$ and $V_{GS} = 2.5$. If $V_{th} = 1\text{ V}$ and $k_n = 1\text{ mA/V}^2$, what is the value of I_D .
(i) 1 mA (iii) 3.5 mA
(ii) 2 mA (iv) 4 mA
- (i) Which component is used to maintain a constant voltage across the load?
(i) P-N junction diode (iii) Schottky diode
(ii) Zener diode (iv) Tunnel diode
- (j) Which configuration of BJT offers the highest current gain?
(i) Common Base (iii) Common Emitter
(ii) Common Collector (iv) Emitter Follower

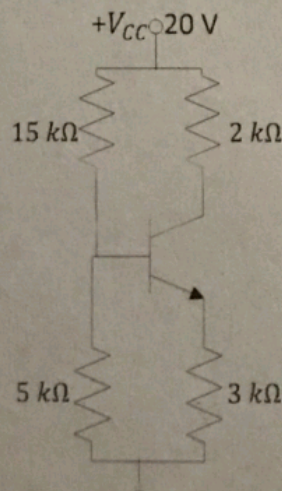
- Q.2 (a) Explain the working of a differential amplifier using BJTs. Derive expressions for differential and common-mode gain. [7]
 (b) Assume the op-amp is ideal, find the output voltage V_{out} . [7]



- Q.3 (a) Design and analyze a Wein Bridge Oscillator using op-amp. Derive the expression for frequency of oscillation. [7]
 (b) Explain the operation of a Zener diode as a voltage regulator. Design a circuit to provide 5V regulated output. [7]
- Q.4 (a) Explain the operation of a hysteretic comparator using op-amp. Draw and analyze the transfer characteristics. [7]
 (b) Assume the op-amp is ideal, find the output voltage V_{out} . [7]



- Q.5 (a) Design and analyze a clamping circuit. Explain how it shifts the DC level of an input waveform. [7]
 (b) Explain the effect of channel length modulation on the performance of a MOSFET amplifier. [7]
- Q.6 (a) Design a triangular wave generator using op-amps. [7]
 (b) For the circuit shown below, draw D.C. load line. Assume β is very large. [7]



- Q.7 (a) Design a first-order low-pass butter worth filter circuit using op-amp and derive the expression of high cut off frequency and transfer function of the filter. [7]
(b) Explain the role of op-amp in waveform shaping circuits with examples. [7]

- Q.8 (a) Derive the expression for ripple voltage and peak inverse voltage in a full-wave rectifier. [7]
(b) Explain the various biasing techniques for a BJT amplifier. Suggest which one is better and why? [7]

Q.9 Write short notes on **any two** of the following:-

[7 x 2 = 14]

- (a) Current mirror using BJT
(b) Trans-conductance of MOSFET
(c) Junction break-down in Zener diodes
(d) high-frequency equivalent circuit of BJT
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